Linear Inequalities

Exam code: 9MA0

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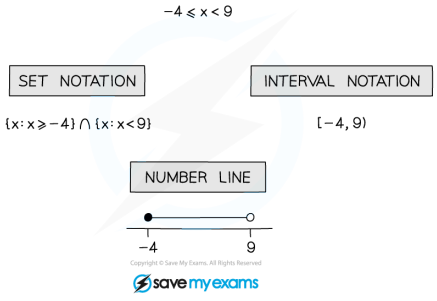
Help me understand

Linear inequalities

What are linear inequalities?

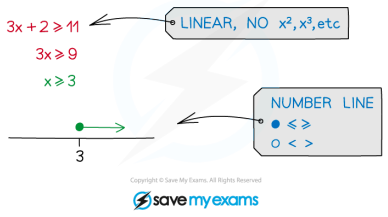
- Linear inequalities are similar to equations but answers take a range of values
- Linear means there will be no terms other than degree 1
  - no squared terms or higher powers, no fractional or negative powers
- Inequalities use the following symbols
  - $>$ Greater than e.g. $5 > 3$
  - $<$ Less than e.g. $-8 < 7$
  - $\geq$ Greater than or equal to
  - $\leq$ Less than or equal to
- Inequalities can be represented in many ways using number lines, set notation and interval notation



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

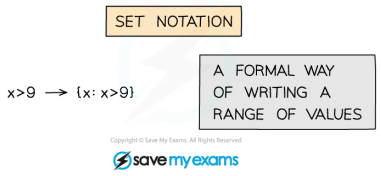
set above a number line

- A filled-in circle or empty circle above a number denotes whether the number is included or not
  - filled in for the greater/less than or equal to symbols
  - empty for the greater/less than symbols
- Arrows show the range of values that are allowed



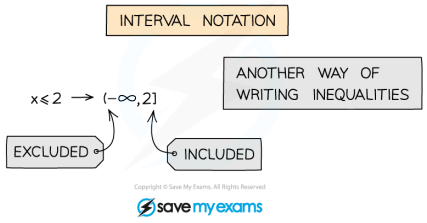
How do I use set notation for inequalities?

- Set notation is a formal way of writing a range of values
- Use of curly brackets { }
- Intersection ∩ and union ∪ may be used
- Not to be confused with interval notation



How do I use interval notation for inequalities?

- Interval notation uses different brackets to indicate whether a number is included or not
- Use of square [] and round () brackets
- [or] mean included
- (or) mean excluded
  - (4,8] means $4 < x \leq 8$
- Note ∞ always uses (or)
- Not to be confused with set notation

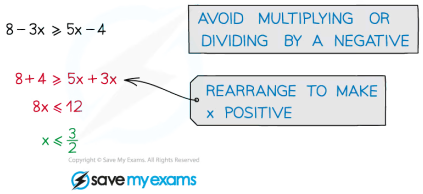


number line

- **writing** and **interpreting** set notation
  - eg $\{x : x > 1\} \cap \{x : x \leq 7\}$ is the same as $1 < x \leq 7$
- **writing** and **interpreting** interval notation
  - eg $[-4, 6)$ is the same as $-4 \leq x < 6$

How do I solve linear inequalities?

- Treat the inequality as an equation and solve
  - avoid multiplying or dividing by a negative
  - if unavoidable, "flip" the inequality sign so $< \rightarrow >$, $\geq \rightarrow \leq$, etc
  - try to rearrange to make the x term positive



Worked Example

Teacher only

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- Fresh, expert-written questions
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